

Mastering the data challenge in AI

Insights from global CIOs and technologists



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Contents

- 3** Foreword from Databricks and SAP
- 4** Acknowledgments
- 5** Introduction
- 6** Bridging the AI-business data divide
- 7** Breaking down data silos
- 9** Elevating data governance
- 10** Conclusion: achieving a unified data view

Foreword from Databricks and SAP

This study reveals that while enterprises are eager to leverage generative artificial intelligence (GenAI) for key business initiatives, they struggle to connect dispersed, siloed business data to AI applications. The findings highlight the need to address fundamental infrastructure challenges and provide a unified view with trusted data, so AI can solve real business problems.

The study also finds that organisations want to build secure and compliant AI systems to drive faster, trusted insights while protecting sensitive data. A unified data landscape facilitates pathways for built-in governance, so organisations have consistent visibility into their AI systems as they evolve.

While achieving these technological advancements is no small feat for organisations, real success with AI lies in unlocking the value of their own proprietary data and grounding AI models in unique business contexts and processes. This allows organisations to build robust AI systems that truly understand the organisation's specific domain and business needs. Databricks and SAP are helping organisations bridge the technological gap and illuminate the path to AI success for building sustainable, data-driven businesses. Together, we invite you to consider the perspectives presented in this report to explore the strategies, obstacles and breakthroughs shaping the future of enterprise AI.



Acknowledgments

The following research team members and specialists contributed to this report. We thank them for their participation.

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We wish to thank the following experts for sharing their insights.

- **Bernd Bucher**, chief information officer, **Novartis**
- **Roman Bugaev**, chief technology officer, **Flo Health**
- **Mohit Kapoor**, group chief technology officer, **Mahindra Group**
- **Ryan Snyder**, senior vice president and chief information officer, **Thermo Fisher Scientific**
- **Sara Vaezy**, chief strategy and digital officer, **Providence**

Introduction



The advent of GenAI has unlocked a wave of spending by corporations across the world. In the US alone, tech companies plan to invest a heady US\$1trn over the next four years to deliver the data centres, chips, energy and infrastructure to support AI, while businesses in every sector are investing in expert staff, computing power and AI models.¹

If companies are to reap the benefits of this breakthrough technology, however, they first need to corral their various data sources into manageable systems so that AI can be put to work on them. Yet obstacles such as data silos and complex data engineering architectures often prevent organisations from achieving a unified view of their data.

To explore how businesses are tackling these challenges, Economist Impact, supported by Databricks and SAP, has independently researched and written this report. The insights are based on cross-tabulated analysis of a subset of survey respondents from our [Unlocking enterprise AI](#) study published in November 2024.² The selected group comprises 450 executives and technologists across the financial services, healthcare and pharmaceuticals, and retail and consumer goods sectors. Insights are also shared from several of our 2024 interviews with chief information officers and chief technology officers at leading organisations across these sectors.

¹ Deloitte, "Can US infrastructure keep up with the AI economy?", 2025, <https://www.deloitte.com/us/en/insights/industry/power-and-utilities/data-center-infrastructure-artificial-intelligence.html>

² Economist Impact's 2024 [Unlocking enterprise AI](#) study combines a global survey of 715 technical executives and 385 data and artificial intelligence (AI) technologists who work across the fields of data engineering, data science and enterprise architecture. It also features insights from interviews with 28 C-suite executives from leading organisations across 11 industries.

Bridging the AI-business data divide

Most organisations struggle to effectively connect AI to their full business data ecosystem

Corporate interest in using AI to draw insights from internal data is clear: in our survey, an overwhelming majority (72%) of firms said they see significant potential in integrating GenAI models with their own proprietary data. Over half of firms are already doing this in some way—either by augmenting external models (57%) or fine-tuning them (52%).³

However, many are struggling to operationalise their full suite of data. **Just 29% of respondents to our survey say their existing architecture can fully connect AI models to relevant business data.** Another 40% can partially connect but still require modifications (see figure 1). Fragmented architectures and technical bottlenecks are common, making it difficult for many organisations to integrate diverse and valuable data sources into a cohesive AI ecosystem.

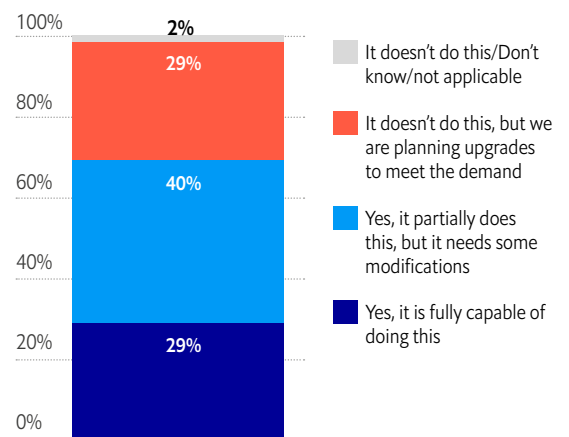
“AI is only as good as the underlying data,” says Bernd Bucher, chief information officer at Novartis, a Swiss pharmaceutical company. “For AI to be effective for a company, it must be built on strong underlying data and data management capabilities.”

To expose the entirety of their companies’ enterprise data to AI, leaders must focus on an interoperable architecture that simplifies data integration and strengthens data management.

Figure 1. Adequacy of current architecture in connecting data to AI

My organisation’s current architecture connects AI to relevant business data. (Select one response)

Respondents in financial services, healthcare and retail



Note: sample size is limited to enterprise architects within the study's 450-respondent subsample, covering the financial services, healthcare and pharmaceuticals, and retail and consumer goods sectors. Interpret with caution. Results are largely similar when all eight industries are included.

Source: Economist Impact 2024. Unlocking enterprise AI — global survey.

³ The preceding insights are based on a subset of executive and technologist respondents to our 2024 “Unlocking enterprise AI” survey, covering the financial services, healthcare and pharmaceuticals, and retail and consumer goods sectors.

Breaking down data silos

Data silos remain a significant organisational barrier for AI

As businesses become increasingly aware of the value that can be harnessed from data, they are gathering more from an ever-increasing array of sources. Public datasets are being combined with proprietary, often unstructured, data and stored across a growing mix of databases, company servers, and public or private cloud networks.

Yet this diversity makes it difficult to quickly assemble relevant information for AI analysis. Efforts to extract and merge different data formats are often hampered by a lack of standardised integration methods, resulting in a slow, resource-intensive process.

Recognising these difficulties, survey respondents identified **dealing with data silos and lack of integration as one of the biggest challenges in managing AI data**, second only to concerns about controlling access (see figure 2).

In a related question about ways to improve data management, three of the top four priorities identified by data engineers emphasise reducing complexity in the data environment.

These include streamlining connections for ingesting data, using a single unified solution instead of multiple tools, and improving visibility into data pipelines to find and fix issues.

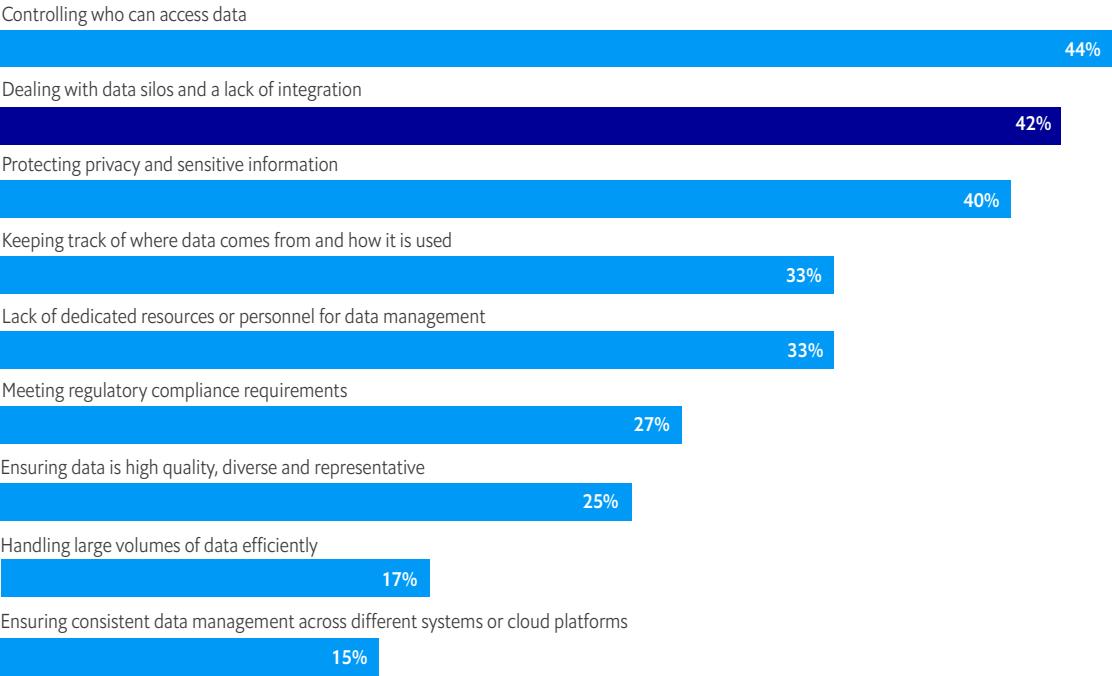
However, data silos are not only a technical problem. They often also reflect deeper organisational issues such as unclear data ownership, restrictive policies or weak collaboration across business lines. To unlock AI's full potential, leaders need to proactively align business and IT on shared data priorities.

Some companies are already showing what's possible. At Mahindra Finance, the financial services arm of Mahindra Group, an AI-powered credit scoring system integrates publicly available landholding data and bureau data with proprietary applicant information. This unified approach, says Mohit Kapoor, the group chief technology officer, has overcome data silos and expanded credit access.

At Thermo Fisher Scientific, Ryan Snyder, senior vice president and chief information officer, believes that once companies create shared datastores and resolve formatting differences, AI will be positioned to operate across industry-wide datasets, enabling breakthroughs in cross-cutting research.



Figure 2. Poor data integration poses widespread challenges for AI
What are the biggest challenges your organisation faces in managing and controlling data for AI applications? (Select up to three)
Respondents in financial services, healthcare and retail



Note: sample size is limited to enterprise architects within the study's 450-responder subsample, covering the financial services, healthcare and pharmaceuticals, and retail and consumer goods sectors. Interpret with caution. Results are largely similar when all eight industries are included.
Source: Economist Impact 2024. Unlocking enterprise AI — global survey.

Elevating data governance

Businesses cite lack of visibility and oversight as a major risk to data integrity

Even with improved data integration, many organisations lack the consistent enterprise-wide data management practices needed to maintain high-quality, trustworthy data. Without consistent monitoring across systems, errors and inconsistencies go undetected—and the promise of unified data begins to unravel.

These observability gaps carry a hefty price: poor-quality data costs the average organisation millions of dollars annually.⁴ This perspective is echoed in our survey: **nearly 80% of respondents cited insufficient observability and monitoring as one of the top three contributors to poor data quality at their organisation** (see figure 3).

A more unified operational view of data through strong governance frameworks makes it easier to identify, debug and resolve data quality issues across disparate environments. It is

therefore critical for organisations to establish a clear strategy for data governance, with a clear mandate for promoting data quality by monitoring compliance with enterprise-wide data management policies.

But improving data oversight doesn't necessarily mean centralising everything. There is a delicate balance to be struck between enabling organisation-wide visibility and safeguarding sensitive information. For companies handling highly regulated or personal data, in particular, **a unified view of data involves implementing intelligent governance with controls and processes to prevent unauthorised access to sensitive data.**

Providence, a US healthcare organisation, for example, implemented an architecture where each piece of data is accessed directly at its source rather than replicated in a single system.

Nearly 80% of respondents cited “insufficient observability and monitoring” as one of the top three contributors to poor data quality at their organisation.

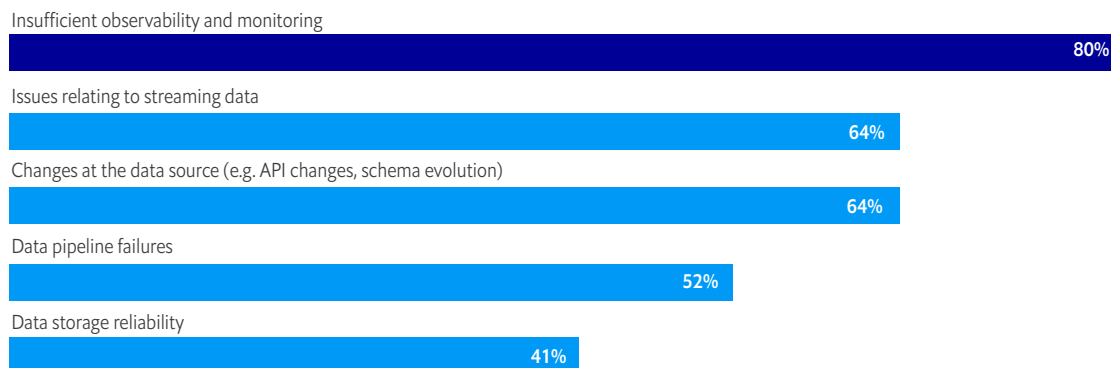
Economist Impact 2024. Unlocking enterprise AI — global survey.

⁴ Gartner, “How to Stop Data Quality Undermining Your Business,” 2018, <https://www.gartner.com/smarterwithgartner/how-to-stop-data-quality-undermining-your-business>

Figure 3. Observability and monitoring gaps stand out as major risks to data quality

The most critical challenges affecting data quality in analytics and AI projects. (Percentage that ranked each challenge among their top three)

Respondents in financial services, healthcare and retail



Note: sample size is limited to data engineers within the study's 450-respondent subsample, covering the financial services, healthcare and pharmaceuticals, and retail and consumer goods sectors. Interpret with caution. Results are largely similar when all eight industries are included.

Source: Economist Impact 2024. Unlocking enterprise AI — global survey.

"When it comes to sensitive medical data, we use a stateless environment," says Sara Vaezy, chief strategy and digital officer at Providence. "We're not storing data in a different warehouse, we're just connecting the dots across existing systems."

Whatever the architecture, the key is to develop clear controls over who can access any given piece of data. This can be achieved with a single tool for data storage as long as datasets are partitioned within the

tool and different access controls are imposed for each of them.

Once access rules are established, high levels of security are required to ensure these are enforced. Providence achieved this by investing in its own cloud environment, enabling the organisation to enforce rules within a company-managed infrastructure—ensuring an overall view of data was possible when required but without compromising security.

Conclusion: achieving a unified data view

A unified view of enterprise data is more than a technical milestone—it is a strategic enabler for unlocking the full potential of AI. When done successfully, the payoff can be significant.

At Providence, for example, Ms Vaezy says: “We’ve developed a very sophisticated data platform that now houses 1.3bn de-identified clinical notes, mostly unstructured data, as well as even more structured data. Our work was supercharged when we brought everything to the cloud and implemented this data platform.”

The path to achieving this level of data integration is rarely straightforward. As Roman Bugaev, chief technology officer at Flo Health, explains, “Our data platform is built on top of various technologies, and we take a best-in-class approach—we cherry-pick technologies from different vendors to build a platform optimised for healthcare.”

Among executives who had achieved a unified view, most described it as a long-term process requiring co-ordination, persistence and sustained

investment. One of the biggest internal hurdles, they said, is simply getting stakeholders to recognise the strategic importance of data management.

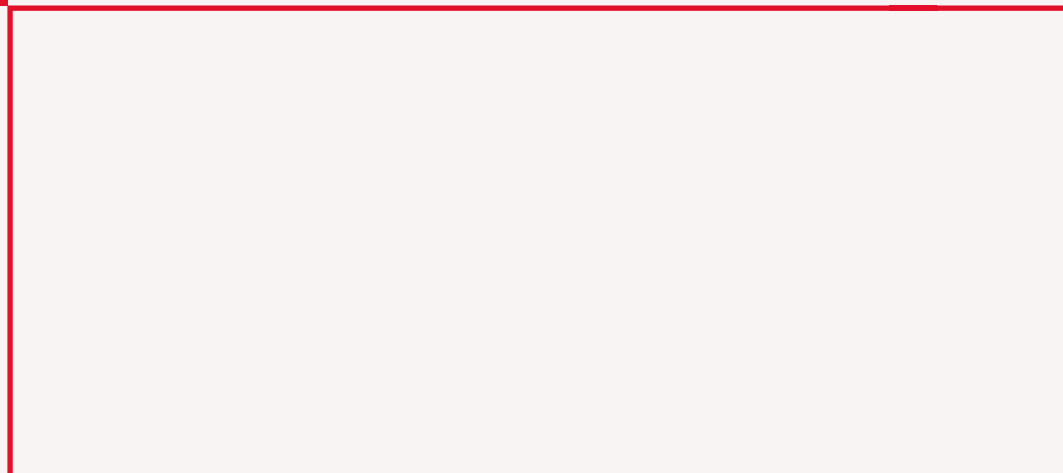
Leaders emphasized that data management and data governance should not feel like a black box. When these processes are made more approachable, stakeholders can more clearly understand the value.

As AI adoption expands across sectors and business functions, the ability to connect and govern data at scale will increasingly define competitive advantage. To achieve this, organisations should:

1. prioritise interoperable architecture;
2. proactively align business and IT on data priorities; and
3. establish a cross-functional data governance team.

Companies that prioritise a unified, enterprise-wide data strategy will be better positioned to translate AI potential into real business value.

While every effort has been taken to verify the accuracy of this information, Economist Impact cannot accept any responsibility or liability for reliance by any person on this report or any of the information, opinions or conclusions set out in this report. The findings and views expressed in the report do not necessarily reflect the views of the sponsor.



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